# NeuroLens

## Watch a brain watch your video.

PREDICTED BRAIN RESPONSE TO VIDEO, SECOND BY SECOND

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# $10k+

to test one ad on a brain scanner

| | | |
|---|---|---|
| fMRI study | **$10 – 15k** | 2–4 weeks |
| NeuroLens | **$1.00** | ≈ 7 min |

Cost per advert. fMRI vs surveys: Venkatraman et al., 2015. Ranges: our report, section 1.3. NeuroLens: $1.00 per 30 s video; about 7 minutes measured for a 30 s clip.

# Ten cuts by lunch.
# No way to judge them.

Generative tools made production cheap. The feed made every second count.



# What already exists: measured, filmed or guessed.

| | | |
|---|---|---|
| Brain scans (fMRI, EEG)<br>research labs, iMotions suites | $10,000 – $15,000 | 2–4 weeks |
| Facial coding<br>Realeyes, iMotions webcam studies | $850 – $4,000 | 3–5 days |
| AI attention maps<br>Neurons AI, Attention Insight | $500 – $1,500 / mo | sub-minute |
| NeuroLens<br>predicted brain response | $1.00 | ≈ 7 min |

## OUR DIFFERENCE

**A brain response, not a gaze or a face**

**Per second, with a no-sound pass**

**Compare any two cuts for $1.00**

Cost and time ranges by category: our report, section 1.3. Vendors are examples of each category, not priced here.

# Under the hood: TRIBE v2.

TRAINED ON REAL BRAINS

**1,000+ hours of fMRI**

From 720 people, on naturalistic video, audio and language.

THREE SENSES IN

**Frames, sound, words**

V-JEPA 2 watches, Wav2Vec-BERT listens, WhisperX and LLaMA 3.2 read the speech.

BRAIN OUT

**20,484 points a second**

Predicted on the cortex for new videos and new people, several-fold better than linear models.

d'Ascoli et al., 2026 (Meta FAIR). CC BY-NC 4.0. It predicts an average viewer.

# Pay as you go. Credit, not cards.

WHO AND HOW MUCH

## $1.00 per started 30 s

For independent creators and small agencies picking one of many AI-made cuts. New users get $10 of credit: ten videos.

WHY PREPAID CREDIT

## Card fees are fixed

About 2.9% + $0.30 a charge: a third of a $1.00 sale, but only 4% of a $20 pack.

Course version: an administrator grants credit; Stripe Checkout belongs to the production design. TRIBE v2 is CC BY-NC, so a paid launch needs a licence.

# What one analysis costs us.

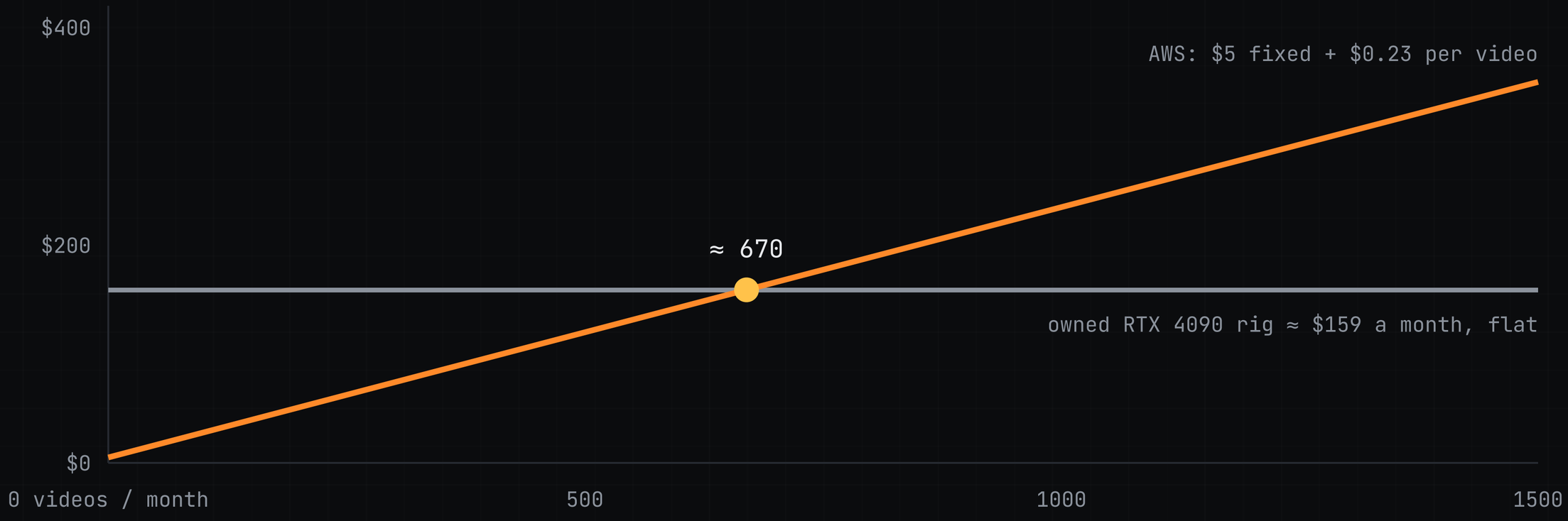
**$0.14–0.23**

our AWS cost per 30 s video, against a $1.00 price

| | |
|---|---|
| GPU: 393 s at $1.22/h (L4) | $0.13 |
| Start-up + idle, per session | 7.5 + 7 min |
| Cost, 3 videos per session | $0.23 |
| Cost, 30 videos per session | $0.14 |
| Fixed cost per month | $5 |
| **Gross margin** | **77 – 86%** |

Report, Table 6. Cost per video = GPU time × hourly price + (start-up + idle) × hourly price ÷ videos per session. Spot GPUs (≈ $0.10 a video) were not available.

# Cloud is cheaper below about 670 videos a month.



# One upload in.
# Seven signals out.

## Your video

MP4, MOV or WebM, up to two minutes

## TRIBE v2

Predicts the brain's response every second

## Seven signals

Seeing: faces, bodies, scenes, visual · Hearing: sound · Understanding: speech, social

## Three outputs

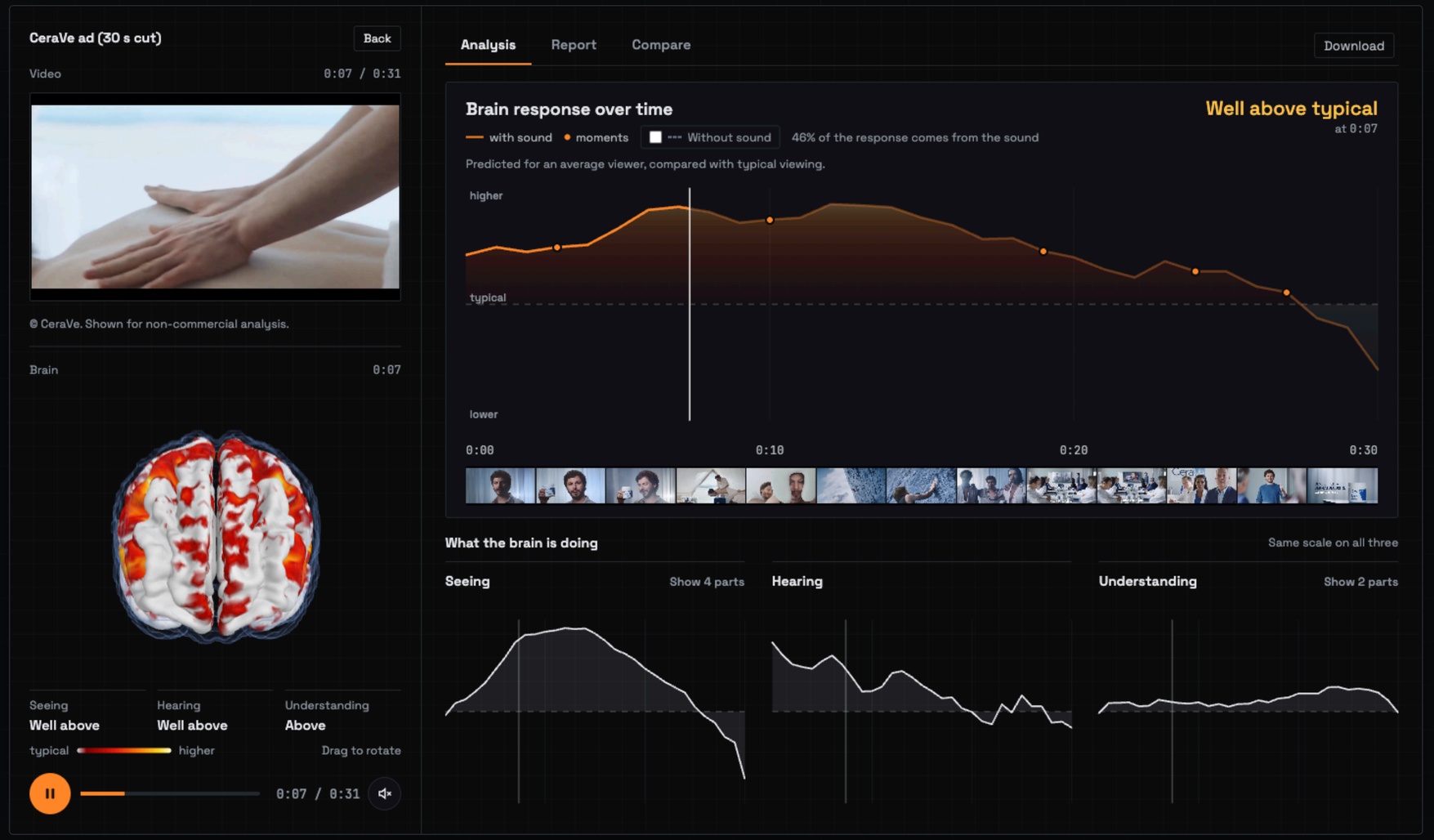
Timeline · 3D brain · AI report

BELOW TYPICAL

0 = TYPICAL FILM & TV VIEWING

ABOVE TYPICAL

# LIVE Live demo: sign in, upload, explore




1. Upload: price first, then confirm
2. Sample result: peak, dip, no-sound toggle
3. AI report and Compare tabs
4. A bad file is rejected, nothing charged

CLICK THE PICTURE TO OPEN THE LIVE SITE · DF6GIRQYYKYFA.CLOUDFRONT.NET

# We pay for GPU minutes only when someone is waiting.




## Check first

Length, size and credit are verified before any GPU wakes.

## Scale to zero

Off after 7 idle minutes: about $5 a month when idle.

## Refund by default

Credit is charged or refunded exactly once. A failed job costs nothing.

# How do we know it works?

THE MODEL

## Validated by Meta

Accurate on new stimuli, tasks and people (d'Ascoli et al., 2026). We did not re-test it against real scans.

OUR PIPELINE

## Same output everywhere

RTX 4090 vs AWS: 11 series, largest difference 0.005, correlation 1.000. 77 of 80 refund tests passed.

OUR USERS

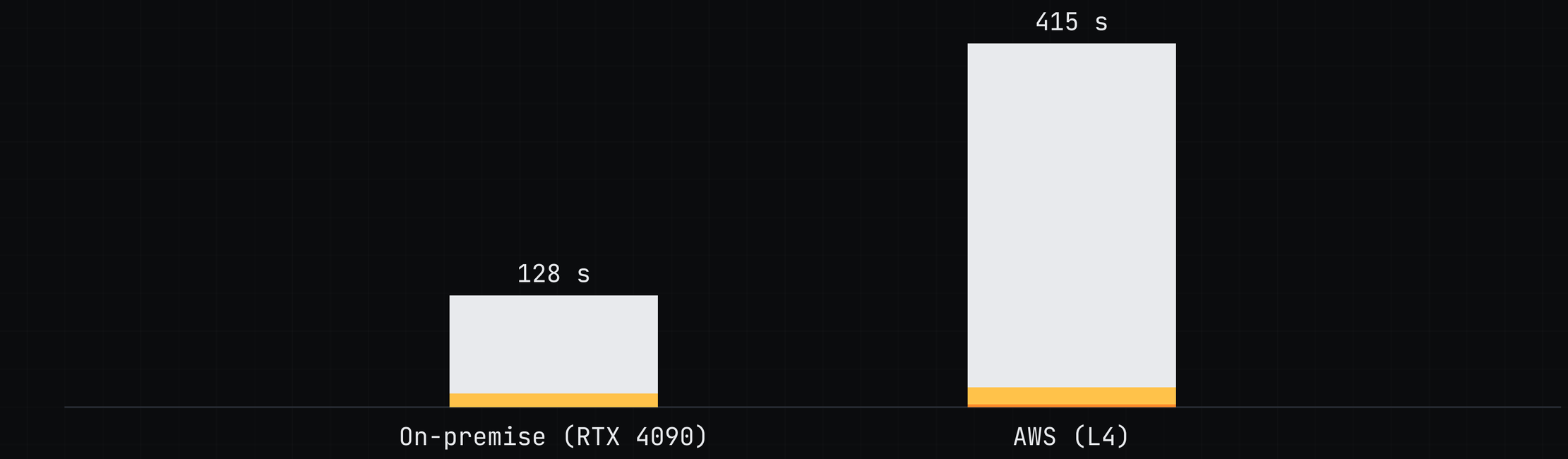
## Usability and value

SUS and a short booth survey: see Usability.

Not validated: that a higher response means more views or sales.

# Where does the time go in one job?

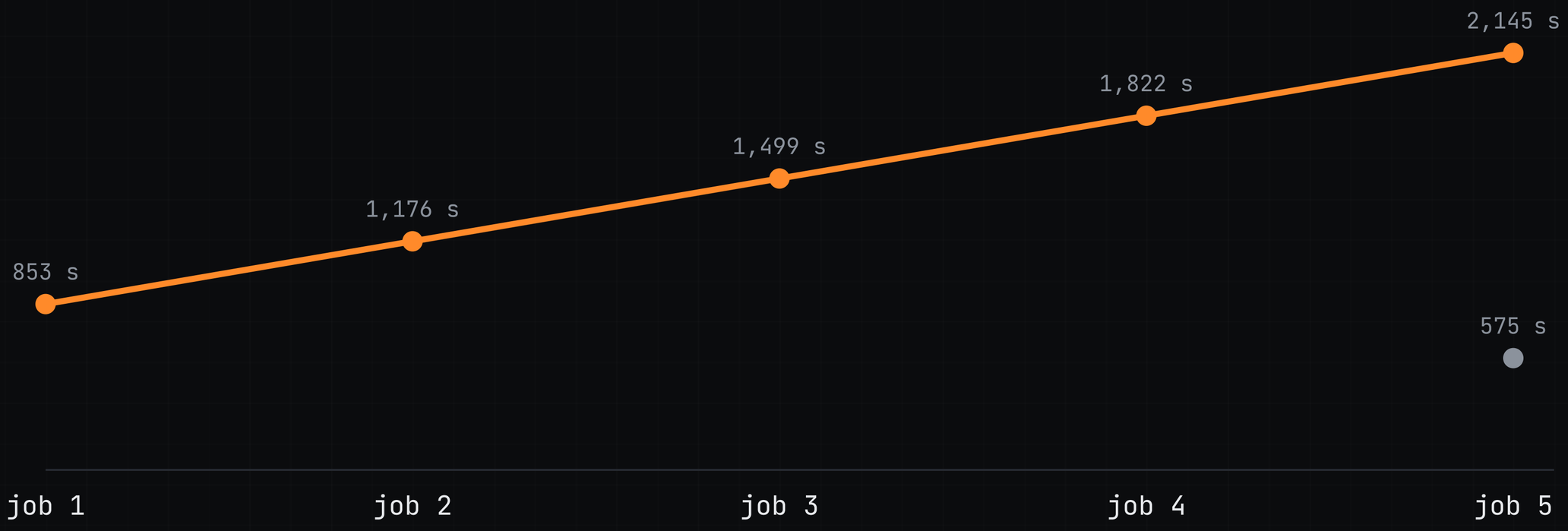
Same code, same 30 s clip, on our own RTX 4090 and on AWS.




GPU TIME PER VIDEO-SECOND: 3.71 S (RTX 4090) VS 10.7 S (A10G) · FIRST VIDEO AFTER 7+ MIN IDLE: 779 S ON AWS

# Five people at once.

Five uploads at once. The endpoint is set to one GPU, so jobs run one after another.




SECONDS UNTIL EACH JOB FINISHES. NO JOB WAS DROPPED. THE AWS RUN GOT AN A10G, AND ITS FIRST JOB APPEARS TO HAVE RESTARTED ONCE.

# Can people use it?

Booth visitors tried a sample or uploaded a clip, then answered the 10-question SUS.

**[TBD]** / 100

SUS SCORE · 68 IS AVERAGE · 80 IS VERY GOOD

[X]% SAW SOMETHING THEY HAD NOT NOTICED
[Y]% WOULD PAY $1.00 FOR A 30 S VIDEO

# What it claims, and what it doesn’t.

## WHAT IT IS

- An average viewer’s predicted brain response
- One scale for every video
- A virtual pre-test for editing

## WHAT IT ISN'T

- A prediction of views or sales
- Instant: first video after a quiet spell ≈ 6 min
- Statistically proven (small test)

## ETHICS

- CC BY-NC: no commercial use without a licence
- No real person is scanned or profiled
- For checking edits, not manipulating people

# Where next: scaling up.

### Spot did not deliver

All 26 Spot launches failed for lack of capacity. Spot averaged only 1–9% below on-demand.

### A smaller GPU fits

The model peaked at 19.9 GB, so a 24 GB L4 or A10G is enough.

### The GPU sets the speed

119 s clip: 491 s on an L40S, 1,540 s on an L4.

Production design (EC2 GPU pool behind a load balancer): $0.10 – $0.20 a video, about $70 a month more fixed cost. Cheaper than SageMaker on the same L40S above about 480–1,440 videos a month. Not built: estimates.

# A $10,000 brain study, in 7 minutes for $1.

Live today: sign-in, upload, prediction, 3D brain, AI report, comparison, refunds.

Next

## Licence

TRIBE v2 is CC BY-NC: a paid launch needs a licence.

Next

## Scale

A GPU pool that grows with the queue: cheaper above about 1,000 videos a month.

Next

## Polish

Fix what our usability study flagged.

# Thank you.
# Questions?

df6girqyykyfa.cloudfront.net

CODE · GITHUB.COM/KELVINLOWHM86/NEUROLENS

# Recorded demo

[Embed the recorded demo of slides 8–11 here]